



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2025-26(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

Class Test (CT)-1

Semester: IV

Programme: B. Tech Information Technology

Course: Database Management System

Time: 1 Hour

Max Marks: 20 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT32402	Course Outcome
CO1	Understand the fundamental of database system and DBMS concepts.
CO2	Identify the concept of conceptual data modeling using the E/R model.
CO3	Illustrate the concept of embedded SQL and its role in application development.
CO4	Analyze the process of database normalization and the steps for achieving 1NF,2NF,3NF,and BCNF
CO5	Compare Various indexing techniques like hash-based indexing,dynamic hashing,and B+trees.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Identify a real-world example where a DBMS is used and enlist its benefit.	CO1	2	2
b.	Discuss the potential consequences of violating referential integrity in a database system.	CO2	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Outline the various components of a DBMS? Explain with neat diagram.	CO1	2	4
b.	Illustrate the three-schema architecture help in database abstraction?	CO1	4	4
c.	Analyze the advantages of the database approach compared to early file systems.	CO1	2	4

Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Express entity relationship model with example	CO2	2	4
b.	Interpret the need of primary key with suitable example	CO2	2	4
c.	Distinguish between strong entity and weak entity.	CO2	2	4

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create